

Illumina PCR-Free DNA Prep

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Version	Enforcement Date
1.0	2024-09-12

REVISIONS FROM THE ORIGINAL VERSION

Version	Summary of modifications	Date and initials

Date: _____

Project: _____

Library prepared by: _____

Purpose: This protocol uses the Illumina PCR-Free DNA Prep kit to convert genomic DNA input from cells, frozen tissue and FFPE into dual-indexed single-stranded DNA libraries for sequencing on Illumina platforms.

Illumina PCR-Free DNA Prep.

Prepwork:

- Sort the kits out as outlined below
- Replenish consumables at all benches
- Prepare batch excel sheet and print MM vols

Kits

Illumina DNA PCR-Free Prep – Purification
Beads + Buffers (Pre-room, Room Temp.)

- ☐ Tagmentation Wash Buffer (TWB)
- ☐ Resuspension Buffer (RSB)
- ☐ Illumina Purification Beads (IPB)
- ☐ Stop Tagment Buffer (ST2)

Illumina DNA PCR-Free Prep.
Tagmentation Beads + Buffer

- ☐ Tagmentation Buffer 1 (TB1)
- ☐ HP3 (2N NaOH)
- ☐ PCR-Free Tagmentation Beads (BLT-PF)
- ☐ Extension Ligation Mix (ELM)

IDT for Illumina RNA UD Indexes –
Ligation (Post-room -20°C)

- ☐ DNA/RNA UD Indexes (UDP0XXX)

Consumables

- ☐ NF water
- ☐ 8-well strips
- ☐ P10, 20, 200, 1000 tips
- ☐ Reagent reservoir
- ☐ 1.5mL & 2mL tubes
- ☐ 15 & 50mL falcons
- ☐ 5, 10, 25mL pipettes
- ☐ 96-well PCR plate, non-skirted
- ☐ PCR plate sealing film
- ☐ 100% EtOH (Pre and post-room)
- ☐ Qubit tubes
- ☐ Kimwipes, gloves, EtOH, RNase Away

Illumina PCR-Free DNA Prep.

Recommended Input:

500 ng genomic DNA extracted from Frozen Tissue.

PRE-PCR ROOM

☐ **Before starting, take out DNA-PCR Free kit and index set from the freezer and keep @ 4°C**

1. Tagment Genomic DNA:

- ☐ RSB (Room Temp, Purification beads and buffers) - Vortex to mix
 - ☐ BLT-PF (-20°C, Tagmentation beads + buffers) – thaw @ 4°C, vortex.
 - ☐ TB1 (-20°C, Tagmentation beads + buffers) – thaw @ 4°C, vortex.
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- ☐ In a PCR plate, add 2-25 uL of extracted genomic DNA, totalling 500 ng.
Bring the volume to 25 uL using RSB.
 - ☐ Preheat thermocycler (TAG)
 - ☐ Vortex and spin BLT-PF.
 - ☐ Add 15 uL BLT-PF to each well sample.
 - ☐ Pipette to mix (40 uL total).
 - ☐ Vortex and spin TB1.
 - ☐ Add 10 uL TB1 to each sample.
 - ☐ Pipette to mix (50 uL total). Seal plate.
 - ☐ Place on thermal cycler and run TAG protocol.
 - ☐ Lid preheated to 100°C
 - ☐ 41°C for 5 min
 - ☐ **Aliquot 12uL/sample ST2 into PCR strip during thermocycler protocol.**
 - ☐ **Immediately proceed to Post Tagmentation Cleanup.**

2. Post-Tagmentation Cleanup

- ST2 (Room Temp, Purification beads and buffers) - Vortex to mix.
- TWB (Room Temp, Purification beads and buffers) - Vortex to mix.

- ☐ Add 10uL ST2 to each sample using a multi-channel pipette.
- ☐ Pipette to mix (60 uL total).
- ☐ Incubate at room temperature for 2 minutes.
- ☐ Place on magnet for ____ minutes until liquid is clear.
- ☐ Remove and discard all supernatant.
- ☐ Remove from the magnet.
- ☐ Add 150 uL TWB.
- ☐ Pipette to mix until beads are fully resuspended.
- ☐ Place on magnet for 5 minutes until liquid is clear.
- ☐ Remove **Illumina IDT UD Indexes from Post-PCR room -20°C.**
- ☐ Preheat thermocycler to **ELM** program.
- ☐ **Proceed to Step 3: Ligate Indexes**

3. Ligate Indexes

- ELM: (-20°C, Tagmentation beads + buffers) – thaw @ 4°C, vortex and aliquot in PCR Strip.
- Illumina IDT Indexes: (-20°C, Post-PCR Room) – Thaw @ RT, vortex, and spin plate.

- ☐ Once liquid is clear, remove and discard supernatant.
- ☐ Remove plate from magnet.
- ☐ Add 45 uL ELM to each sample.
- ☐ Pipette to mix.
- ☐ Pierce the foil of the intended indexes using 200 uL multi-channel.
- ☐ Add 5 uL index adapters to the associated samples.
- ☐ Pipette to mix, seal the plate.
- ☐ Place plate on thermocycler and run **ELM Protocol.**
 - Lid preheated to 100°C
 - 37°C for 5 minutes
 - 50°C for 5 minutes.

- ☐ During incubation, **prepare diluted HP3.**
 - HP3: 6ul/sample
 - Nuclease-Free Water: 54uL/sample
 - Aliquot in PCR Strip

- ☐ Remove plate from thermocycler and place on magnet until liquid is clear.
- ☐ Remove and discard all supernatant.
- ☐ Remove from magnet.
- ☐ Add 75 uL TWB.
- ☐ Pipette to mix until beads are fully resuspended.
- ☐ Place on magnet until liquid is clear.
- ☐ Remove and discard supernatant.
- ☐ Seal plate and spin briefly.
- ☐ Place on magnetic stand.
- ☐ Without disturbing the pellet, remove any residual supernatant.
- ☐ Remove from magnetic stand, Add 45 uL diluted HP3 to each sample.
- ☐ Pipette to mix, incubate @ RT for 2 minutes.
- ☐ **Proceed immediately to Step 4: Clean up Libraries.**

4. Clean Up Libraries

- ☐ Prepare 80% EtOH (500 μ L / sample)
 - 400 μ L EtOH 100%/Sample
 - 100 μ L Nuclease-Free H₂O/Sample
- ☐ Take out ssDNA Qubit kit for quantification.
- ☐ Vortex Illumina Purification Beads (IPB)
- ☐ Add 36 μ L IPB to each well containing sample + BLT-PF.
- ☐ Pipette to mix.
- ☐ Incubate at room temp for 2 minutes.
- ☐ Place samples on magnet for 5 min or until solution appears clear.
- ☐ Without disturbing the pellet, transfer 76 μ L supernatant to new wells.
- ☐ Add 42 μ L IPB to each sample.
- ☐ Pipette to mix until the beads are fully resuspended. Incubate @ RT for 2 minutes.
- ☐ Place on magnetic stand until liquid is clear.
- ☐ Going one column at a time:
 - Remove and discard the supernatant
 - Immediately add 180 μ L of fresh 80% EtOH
 - Proceed to next column and repeat.
- ☐ **Repeat this procedure for a 2nd bead wash.**
- ☐ Air dry: Start the timer for 2 min
 - Remove and discard the supernatant from all wells, start 2 minute timer.
 - After 1 minute, remove residual liquid with a 10 μ L pipette
 - Allow the pellet to dry, being careful not to over-dry (if the pellet is cracking, then it is too dry). Pellet should appear opaque and slightly shiny.
 - Remove the plate from the magnetic stand
 - Add 22 μ L of RSB directly on the beads
- ☐ Resuspend the pellet thoroughly.
- ☐ Incubate for 2 minutes at room temperature
- ☐ Place on magnet until liquid is clear.
- ☐ Transfer 20 μ L of supernatant to new wells
- ☐ Quick spin
- ☐ Proceed to **Step 5: Quantify and Pool Libraries.**

5. Quantify and Pool Libraries

- ☐ Bring Qubit ssDNA kit to room temp 30 min (**Step 4**)

- ☐ Mix Libraries by shaking at 1800 rpm for 1 min then quickspin, before doing QC.

Qubit

- ☐ Label Qubit tubes (# samples + 2 standards)
 - ☐ Vortex Dye, Standard 1 and Standard 2, then quickspin
 - ☐ Prepare MasterMix (# samples + 2 standards + extra) by diluting Dye 1/200 using Buffer
 - ☐ Follow table taped to MasterMix bench
 - ☐ Add MasterMix to Qubit tubes:
 - ☐ 190µl for Standard (MasterMix bench)
 - ☐ 199µl for Samples (MasterMix bench)
 - ☐ Add Standard/Sample to Qubit tubes:
 - ☐ 10µl for Standard (Lib Prep bench)
 - ☐ 1µl for Samples (Lib Prep bench)
 - ☐ Vortex Qubit tubes
 - ☐ Incubate for 2 min at room temp
 - ☐ Read
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- ☐ Add lib prep plate to plate inventory in the LIMS and update the index inventory.
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- ☐ Update Global Sheet